

Equilibrio General Computable -- Maestría en Economía UNLP

Simulaciones con Modelo Equilibrio General Computable para Economía Abierta y Pequeña: ToT

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Introducción

- En esta presentación veremos, paso a paso, cómo definir y analizar simulaciones del modelo de economía abierta y pequeña (SOE; Static Open Economy).
- In order to follow the exercises, you should have successfully installed
 - GAMS
 - Archetype 2015 dataset

Elasticidades

- In these exercises, we use the following elasticities – consistent with the literature:
 - factor substitution = 1 prv, 1 gov
 - σ_q (Armington) = 2 prv, 2 gov
 - σ_x (CET) = 2 prv, 2 gov
 - sin embargo, elast para gov irrelevantes
- Additional exercise: assess the sensitivity of results with respect to values assigned to these three elasticities.

Definición de los Escenarios: Shocks

- **pweprv15** = aumento 15% precio mundial exportación c-prv
- **pwmprv25** = disminución 25% precio mundial importación c-prv
- **pwprv10** = aumento 10% precio mundial exportación e importación c-prv
- En ambos casos, los precios mundiales están expresados en moneda del resto del mundo y los cambios de precios son respecto de la base.

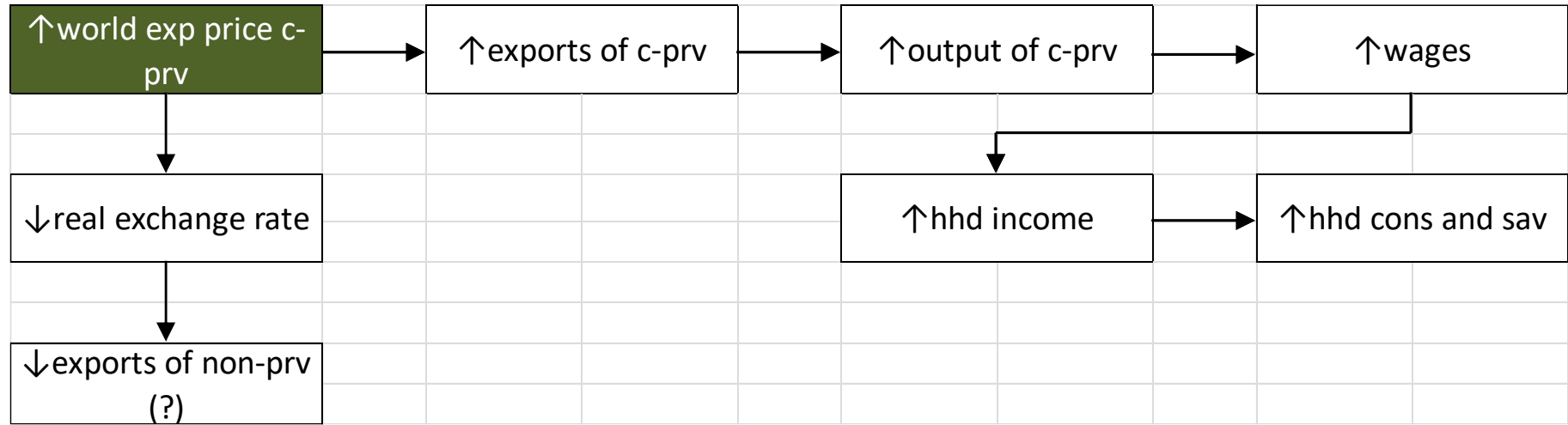
Definición de los Escenarios: Mercados Factoriales

- Trabajo = pleno empleo; movilidad perfecta entre sectores
- Capital = pleno empleo; específico

Definición de los Escenarios: Cierre Macro

- **Gobierno** = **ahorro del gobierno** es la variable que equilibra el presupuesto público.
- **Ahorro-Inversión** = la **inversión** ajusta endógenamente para igualar ahorro e inversión.
- **Sector Externo** = el tipo de cambio ajusta endógenamente para igualar entradas y salidas de divisas.

Canales de Transmisión pweprv



Canales de Transmisión pwmpv

- ???

Año Base: PIB

	Nominal	NomShare
Item	billion US\$	%
Absorption	423.9	105.6
PrvCon	324.5	80.8
GovCon	47.1	11.7
FixInv	77.2	19.2
StockChange	-24.8	-6.2
Exports	79.3	19.8
Imports	-101.7	-25.3
GDPMP	401.6	100.0
GDPMP_alt	401.6	100.0
NetIndTax	30.0	7.5
GDPFC	371.5	92.5

Año Base: Balanza de Pagos

Item	Nominal	GDPshr	NomFCU
In-Exports	79.3	19.8	79.3
In-Trnsfr-NoGov	9.3	2.3	9.3
In-Trnsfr-Gov	3.1	0.8	3.1
In-FacInc	-9.5	-2.4	-9.5
In-RowSav	19.5	4.9	19.5
In-Total	101.7	25.3	101.7
Out-Imports	101.7	25.3	101.7
Out-Total	101.7	25.3	101.7

Año Base: Presupuesto Gobierno

Item	Nominal	GDPshr
Inc-DirTax	16.0	4.0
Inc-ActTax	1.3	0.3
Inc-ComTax	22.8	5.7
Inc-ImpTax	5.8	1.4
Inc-ExpTax	0.1	0.0
Inc-TrnsfrFor	3.1	0.8
Inc-Total	49.1	12.2
Spnd-GovCon	47.1	11.7
Spnd-TrnsfrDom	-1.0	-0.3
Spnd-GovSav	3.0	0.8
Spnd-Total	49.1	12.2

Aspectos Clave Año Base: Estructura Sectorial (%)

Commodity	VAshr	PRDshr	EMPshr	EXPshr	EXP- OUTshr	IMPshr	IMP- DEMshr
c-prv	96.4	92.9	93.6	100.0	12.9	100.0	16.1
c-gov	3.6	7.1	6.4	0.0	0.0	0.0	0.0
total	100.0	100.0	100.0	100.0	12.0	100.0	15.1

Aspectos Clave Año Base: Intensidad de Uso Factorial (%)

Activity	lab	cap	total
a-prv	55.1	44.9	100.0
a-gov	100.0		100.0
total	56.7	43.3	100.0

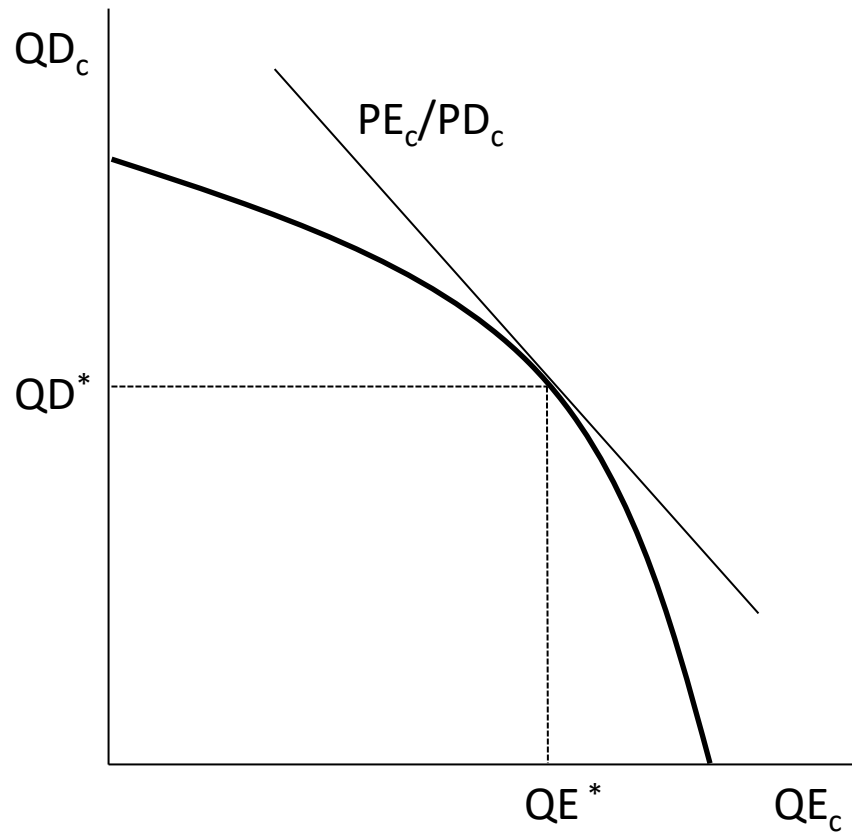
Aspectos Clave Año Base: Estructura Demanda (%)

Commodity	Interm Con	PrvCon	GovCon	FixInv	StockCh ange	Exports	TotDem
c-prv	38.8	43.5		10.4	-3.3	10.6	100.0
c-gov			100.0				100.0
total	36.5	41.0	5.9	9.7	-3.1	10.0	100.0

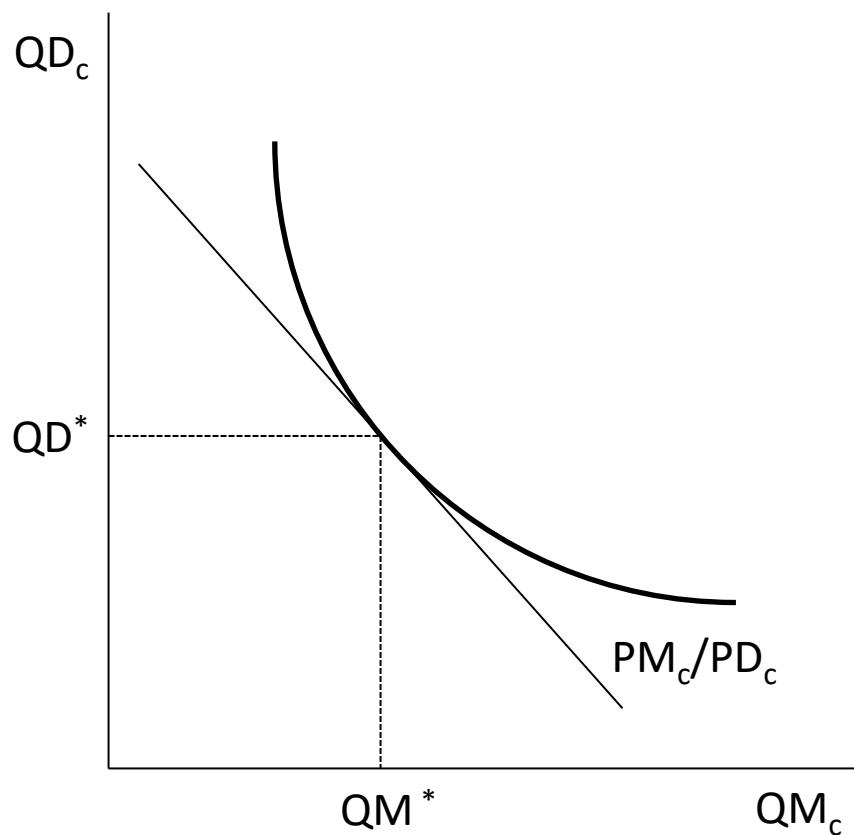
Aspectos Clave Año Base: Estructura Ingresos Instituciones (%)

Item	hhd	gov	row
tot-tax		93.7	
trgov	-0.3		
trrow	2.5	6.3	
imports-2			100.0
lab	56.8		
cap	41.0		
total	100.0	100.0	100.0

Ventas Nacionales versus Exportaciones – cont.



Compras Nacionales versus Importaciones – cont.



Ecuaciones y Variables Clave: Precios de Importación y Exportación

$$PE_c = (1 - te_c)EXR \cdot pwe_c \quad c \in \mathcal{C}$$

$$\frac{QE_c}{QD_c} = \left(\frac{PE_c}{PD_c} \frac{\delta_c^{ds}}{\delta_c^e} \right)^{\frac{1}{\rho_c^x - 1}} \quad c \in \mathcal{C}$$

$$PM_c = (1 + tm_c)EXR \cdot pwm_c \quad c \in \mathcal{C}$$

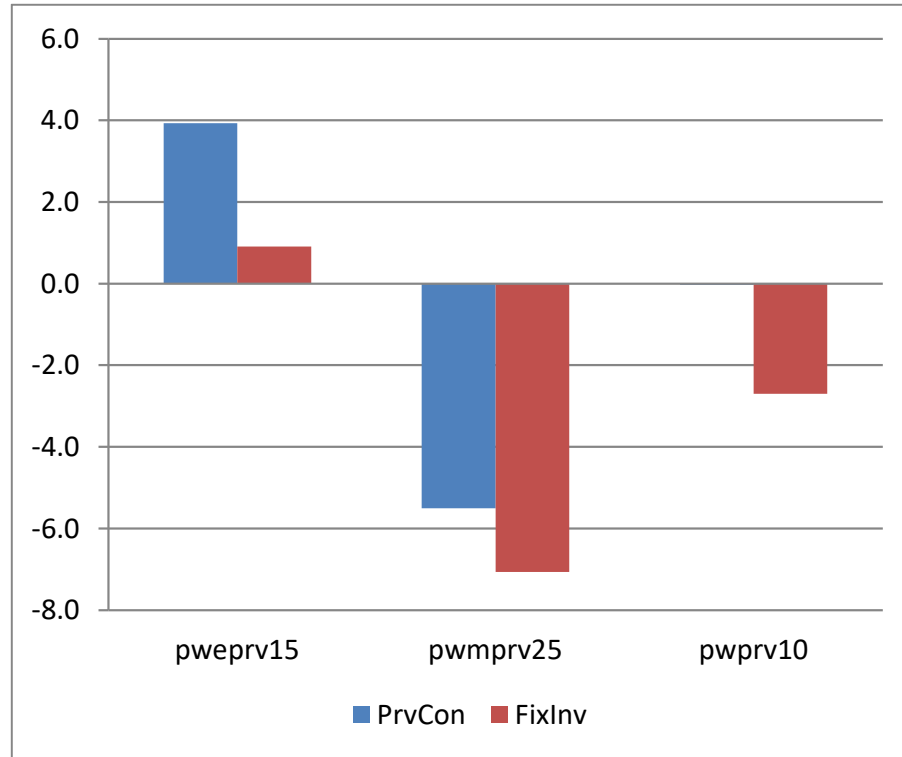
$$\frac{QM_c}{QD_c} = \left(\frac{PD_c}{PM_c} \frac{\delta_c^m}{\delta_c^{dd}} \right)^{\frac{1}{1 + \rho_c^q}} \quad c \in \mathcal{C}$$

Ecuaciones y Variables Clave: Balanza de Pagos (en FCU)

$$\sum_{c \in C} pwe_c \cdot QE_c + trnsfr_{hhd,row} + trnsfr_{gov,row} + trnsfr_{lab,row} \\ + trnsfr_{cap,row} + savf = \sum_{c \in C} pwm_c \cdot QM_c$$

- En palabras,
 - BoP (balance of payments) is in foreign currency.
 - left-hand side: inflows of foreign exchange.
 - right-hand side: outflows of foreign exchange.

Consumo Privado e Inversión – cambio % respecto base



Question: why should we analyze private consumption and fixed investment?

Abbreviations in Macro Reports

- PrvCon = Private consumption
- FixInv = Fixed investment
- StockChange = Stock change
- GovCon = Government consumption
- GDPMP = GDP at market prices
- NetIndTax = Net indirect taxes
- GDPFC = GDP at factor cost

Abbreviations in BoP Reports

- In-Exports = Exports
- In-Trnsfr-NoGov = Transfers to non-gov from RoW
- In-Trnsfr-Gov = Transfers to gov from RoW
- In-FacInc = Factor income from RoW
- In-RowSav = RoW savings (current account deficit)
- In-Total = Total inflows of foreign exchange
- Out-Imports = Imports
- Out-Trnsfr-NoGov = Transfers from non-gov to RoW
- Out-Trnsfr-Gov = Transfers from gov to RoW
- Out-FacInc = Factor income to RoW
- Out-Total = Total outflows of foreign exchange

Abbreviations in Fiscal Reports

- Inc-DirTax = Direct tax
- Inc-FacTax = Factor income tax
- Inc-ActTax = Output tax
- Inc-ComTax = Commodity tax
- Inc-ImpTax = Import tax
- Inc-TrnsfrFor = Transfers from RoW
- Inc-FacInc = Factor income
- Inc-Total = Total receipts
- Spnd-GovCon = Gov consumption
- Spnd-TrnsfrDom = Transfers to dom inst
- Spnd-TrnsfrFor = Transfers to RoW
- Spnd-GovSav = Government savings
- Spnd-Total = Total spending

Macro Results (real quantities)

(% change from base)

	base	pweprv15	pwmprv25	pwprv10
Absorption	423.94	3.17	-5.50	-0.52
PrvCon	324.47	3.93	-5.50	-0.03
GovCon	47.12			
FixInv	77.16	0.91	-7.06	-2.70
StockChange	-24.80			
Exports	79.33	8.27	-10.36	0.97
Imports	-101.71	19.11	-26.46	-1.24
GDPMP	401.56	0.14	-1.15	-0.04
NetIndTax	30.04	5.37	-7.58	-0.49
GDPFC	371.52			
REXR	1.00	-8.91	-6.09	-8.59
base = billion US\$				

Sectoral Results: Real Value Added (% change from base)

Sectoral Results: Real exports (% change from base)

	base	pweprv15	pwmprv25	pwprv10
c-prv	79	8.27	-10.36	0.97
total	79	8.27	-10.36	0.97

Sectoral Results: Real imports (% change from base)

	base	pweprv15	pwmprv25	pwprv10
c-prv	101.71	19.11	-26.46	-1.24
total	101.71	19.11	-26.46	-1.24

Sectoral Results: Real Domestic Sales (% change from base)

	base	pweprv15	pwmprv25	pwprv10
c-prv	535	-1.25	1.49	-0.14
c-gov	47			
total	583	-1.15	1.37	-0.13

Real Wages (rents) by Factor (% change from base)

	base	pweprv15	pwmprv25	pwprv10
tot-lab	0.97	3.91	-5.49	-0.04
lab	0.97	3.91	-5.49	-0.04
cap	0.97	3.91	-5.49	-0.04

Government Budget (% of GDP)

	base	pweprv15	pwmprv25	pwprv10
Inc-DirTax	15.97	16.60	15.09	15.96
Inc-ActTax	1.26	1.29	1.22	1.26
Inc-ComTax	22.84	23.30	22.04	22.76
Inc-ImpTax	5.80	6.40	4.89	5.76
Inc-ExpTax	0.14	0.16	0.12	0.14
Inc-TrnsfrFor	3.09	2.86	2.84	2.83
Inc-Total	49.11	50.61	46.19	48.71
Spnd-GovCon	47.12	47.64	46.38	47.11
Spnd-TrnsfrDom	-1.05	-1.05	-1.05	-1.05
Spnd-GovSav	3.03	4.01	0.85	2.65
Spnd-Total	49.11	50.61	46.19	48.71

Ejercicios Adicionales

- Ahora, correr las mismas simulaciones pero suponiendo que
 - (a) las elasticidades Armington son 0.5 en lugar de 2 y las elasticidades CET son 0.5 en lugar de 2
 - (b) el factor capital es móvil entre sectores

Macro Results with Alternative Trade Elasticities: % change from base

	base	pweprv15	pwmprv25	pwprv10
Absorption	423.94	2.79	-6.11	-0.52
PrvCon	324.47	4.20	-7.57	-0.14
GovCon	47.12			
FixInv	77.16	-2.30	-1.74	-2.27
StockChange	-24.80			
Exports	79.33	-1.93	5.43	0.97
Imports	-101.71	9.96	-16.61	-1.25
GDPMP	401.56	0.04	-1.17	-0.04
NetIndTax	30.04	3.36	-5.84	-0.49
GDPFC	371.52			
REXR	1.00	-16.72	12.71	-7.08
base = billion US\$				

En este caso, debemos recalibrar el modelo; no es necesario redefinir los escenarios de simulación.

Exports and Imports with Alternative Trade Elasticities

pweprv: % change from base

